Software Requirements Specification (SRS)

Fleet Management System

Log company vehicles, track maintenance, monitor fuel costs and schedule drivers

Document	Version	Status
Software Requirements Specification (SRS)	1.0	Draft for review

1. Introduction

1.1 Purpose

This document specifies the requirements of the Fleet Management System (FMS), a Java-based web application that helps a company manage its vehicles, maintenance records, fuel expenses and driver schedules. It is intended for developers, testers, the project guide and end users.

1.2 Scope

FMS provides a central record of all company vehicles and drivers, tracks service and repair history with reminders for upcoming maintenance, records fuel purchases and calculates fuel cost and mileage, and assigns drivers to vehicles and trips through a schedule that avoids conflicts. It also produces summary reports. Real-time GPS tracking and payroll are outside the scope of this version.

1.3 Definitions and Abbreviations

Term	Meaning
FMS	Fleet Management System
Odometer	Distance reading of a vehicle in km
Mileage / Fuel efficiency	Distance travelled per litre of fuel (km/l)
Preventive maintenance	Scheduled service based on date or odometer interval
Trip	A driver-vehicle assignment for a time period and route
SRS / SDD	Software Requirements Specification / Software Design Description
CRUD	Create, Read, Update, Delete

1.4 References

- IEEE Std 830-1998 / ISO/IEC/IEEE 29148
- Companion document: Software Design Description (SDD) for FMS

2. Overall Description

2.1 Product Perspective

FMS is a standalone web application with a browser-based frontend, a Java (Spring Boot) backend and a relational database. It does not depend on external systems; an optional e-mail service is used for reminders.

2.2 Product Functions (Summary)

- Vehicle management: register vehicles, store details, documents and status.
- Driver management: register drivers, licence details and availability.
- Maintenance tracking: log services and repairs, set service intervals, get due-date reminders.
- Fuel monitoring: record refuelling, calculate cost per trip, per vehicle and mileage.
- Driver scheduling: assign drivers to vehicles and trips, with conflict detection.
- Dashboard and reports: costs, maintenance status, vehicle utilisation, expiring documents.

2.3 User Classes

User	Description	Privileges
Administrator	Manages users, master data and settings	Full access
Fleet Manager	Manages vehicles, maintenance, fuel and schedules; views reports	Create, edit, approve, report
Driver	Views own schedule, logs fuel and trip odometer readings	Limited: own data only

2.4 Operating Environment

- Server: Java 17+, Spring Boot, MySQL 8 or PostgreSQL 14+.
- Client: modern web browsers on desktop, tablet or mobile.

2.5 Constraints and Assumptions

- Currency and distance units are configurable (default: INR, km, litres).
- Odometer readings are entered manually by drivers or managers.
- Users have an internet or intranet connection to reach the server.

3. Functional Requirements

ID	Requirement	Priority
FR-01	The system shall provide login with role-based access (Admin, Manager, Driver).	High
FR-02	The system shall allow adding, editing, viewing and deactivating vehicles with registration number, make, model, year, type, fuel type, capacity, current odometer and status.	High
FR-03	The system shall track vehicle documents (insurance, permit, pollution certificate, registration) with expiry dates.	Medium
FR-04	The system shall allow adding, editing and deactivating drivers with licence number, licence class, licence expiry and contact details.	High
FR-05	The system shall record maintenance entries: vehicle, type (service, repair, tyre, inspection), date, odometer, vendor, description, cost and parts.	High
FR-06	The system shall let a manager define service intervals (by km or days) per vehicle and show upcoming and overdue maintenance.	High
FR-07	The system shall send reminders (on dashboard and optionally by e-mail) for due maintenance and expiring documents or licences.	Medium
FR-08	The system shall record fuel entries: vehicle, driver, date, litres, price per litre, total cost, odometer and station.	High
FR-09	The system shall calculate mileage (km/l) and cost per km for each vehicle and period.	High
FR-10	The system shall flag abnormal fuel entries (e.g. litres above tank capacity, mileage far below average).	Medium
FR-11	The system shall allow a manager to create trips/schedules assigning a driver and a vehicle for a date-time range, with source, destination and purpose.	High
FR-12	The system shall prevent double booking of a driver or vehicle, and block assignment of vehicles under maintenance or drivers with expired licence.	High
FR-13	The system shall let drivers view their upcoming schedule and mark trips as started or completed with end odometer.	Medium
FR-14	The dashboard shall show total vehicles by status, upcoming maintenance, monthly fuel cost, maintenance cost and today's schedule.	High

ID	Requirement	Priority
FR-15	The system shall generate reports (fleet summary, vehicle cost history, fuel consumption, maintenance history, driver utilisation) and export to PDF and CSV.	Medium
FR-16	The system shall support search, filter and sort on all list screens.	Low
FR-17	The system shall keep an audit log of create, update and delete actions.	Low

3.1 Key Calculations

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Mileage (km/l) = (Odometer_current - Odometer_previous) / Litres filled
Fuel cost per km = Total fuel cost / Distance travelled
Total cost of ownership (period) = Fuel cost + Maintenance cost
Next service due (km) = Last service odometer + Service interval (km)
Driver utilisation (%) = Hours on trips / Available hours x 100
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4. External Interface Requirements

4.1 User Interface

- Responsive web UI with sidebar navigation: Dashboard, Vehicles, Drivers, Maintenance, Fuel, Schedule, Reports, Admin.
- Forms with validation messages; tables with search, filter and pagination.
- Colour-coded status for vehicles (Available, On trip, Under maintenance, Inactive) and due dates (green, amber, red).

4.2 Software and Communication Interfaces

- REST/JSON API between frontend and backend over HTTP/HTTPS.
- JDBC/JPA connection to the relational database.
- SMTP for optional e-mail reminders.

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	List and dashboard pages shall load within 3 seconds with up to 10,000 records.
NFR-02	Security	Passwords hashed (BCrypt); role-based authorisation; protection against SQL injection and XSS; HTTPS in deployment.
NFR-03	Reliability	Data changes are transactional; daily database backup.
NFR-04	Usability	Common tasks (add fuel entry, assign driver) in no more than three screens or clicks.
NFR-05	Scalability	Support at least 500 vehicles and 1,000 users.
NFR-06	Maintainability	Layered architecture, documented code, unit-test coverage of at least 70% on service layer.
NFR-07	Portability	Runs on Windows and Linux; browser-independent.
NFR-08	Data integrity	Referential integrity enforced; no deletion of vehicles or drivers with history (soft delete only).

6. Data Requirements

Main data items and basic validation rules:

Entity	Key fields	Rules
Vehicle	Reg. no., make, model, year, fuel type, tank capacity, odometer, status	Reg. no. unique; odometer never decreases
Driver	Name, phone, licence no., class, expiry, status	Licence no. unique; expiry date must be in future to assign
Maintenance	Vehicle, type, date, odometer, vendor, cost, next due	Cost >= 0; odometer >= last reading
Fuel entry	Vehicle, driver, date, litres, price/litre, total, odometer	Litres > 0; litres <= tank capacity
Trip	Vehicle, driver, start, end, route, purpose, status	No overlap for same driver or vehicle
Document	Vehicle, type, number, expiry	Expiry date required

7. Use Cases

Use case	Actor	Main flow
UC-01 Register vehicle	Manager	Open Vehicles, click Add, enter details, save
UC-02 Log maintenance	Manager	Select vehicle, enter service details and cost, system updates next due date
UC-03 Record fuel	Driver/Manager	Select vehicle, enter litres, price and odometer; system computes cost and mileage
UC-04 Schedule driver	Manager	Choose vehicle, driver and time; system checks conflicts and confirms
UC-05 View schedule	Driver	Log in, view assigned trips, mark started or completed
UC-06 Review reports	Manager	Select report type and period, view and export

8. Acceptance Criteria

- A vehicle, driver, maintenance record, fuel entry and trip can each be created, edited and viewed correctly.
- Mileage and cost per km match hand calculations on sample data.
- The system rejects overlapping bookings and assignments of vehicles under maintenance or drivers with expired licences.
- Overdue maintenance and expiring documents appear on the dashboard.
- Reports export correctly as PDF and CSV.